Set notation and sample space

1 Heads and Tails

2

a $\{1, 2, 3, 4, 5, 6\}$ **b** $\{1, 2\}$ **c** $\{3, 6\}$ **d** $\{2, 4, 6\}$

3

a $1, 2, 3, 4, 5, 6$ **b** $1, 2$

4

a $\{2, 4, 6, 8, 10, 12, 14\}$ **b** $\{-7, -6, -5, -4\}$

c $\{1, 3, 5, 7, 9, 11\}$ **d** $\left\{1, \frac{1}{10}, \frac{1}{100}, \frac{1}{1000}, \frac{1}{10\,000}, \frac{1}{100\,000}\right\}$

e $\{2, 4, 8, 16, 32, 64\}$

5

a $\{1, 2, 4, 8, 16\}$ **b** $\{1, 2, 4, 8\}$

Set notation (Extended)

6 $\{x : x \text{ is an even number and } 2 \leq x \leq 8\}$

7

a True **b** True **c** False **d** True

8

a $\subseteq$ **b** $\subseteq$ **c** $\nsubseteq$ **d** $\nsubseteq$

9

a $\in$ **b** $\notin$

10

a $\{g, a, l, o, p\}$ **b** Yes

c $\{p, a, r, l, e, o, g, m\}$

11 The set of all the elements in the universal set that are not in set A.

12

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a $A = \{0, 1, 2, 3, 4, 5\}$ **b** Yes

13
a 11 **b** 12 **c** 7 **d** 16

14
a No **b** Yes **c** No **d** Yes

15
a Because there is only one point that satisfies both equations, and it occurs at the point of intersection of the lines.

b $(10, 26)$

Operations on sets

16
a The set containing all the elements that are members of set A or of set B or of both sets.

b The set containing all the elements that are common to both set A and set B.

17
a Yes **b** $\{\text{cauliflowers}\}$

18 $\{3, 7\}$

19
a $\{0, 4, 16, 36, 64\}$ **b** $\{5, 10, 15, 20, 25, 30, 35, 40, 45\}$

20
a $\{1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36\}$ **b** $\{1, 2, 3, 4, 6, 12\}$

21
a $\{1, 2, 3, 4, 5, 6\}$ **b** $\{1, 2, 3, 4, 5, 6, 7, 8\}$ **c** $\{1, 2, 3, 4, 5, 6\}$ **d** $\{1, 2, 3, 4, 5, 6, 7, 8\}$

Complement of sets (Extended)

22

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a

i $\{\text{roses, daisies}\}$

ii $\{\text{lillies}\}$

iii $\{\text{roses, daisies, lillies, sunflowers}\}$

iv $\{\text{roses, daisies}\}$

b Yes

23

a Yes

b 5

c $\{7, 13, 12, 26\}$

d $U = \{7, 8, 9, 12, 13, 21, 26, 28, 30\}$

e The empty set $\emptyset$.

24

a $\{5, 6, 7\}$

b $\emptyset$

25

a People who do not like softball.

b People who like either football, softball, or swimming.

26

a Yes

b No

c $\{-5, 3\}$

d $\{11\}$

e 19

f 16

Venn diagrams

27

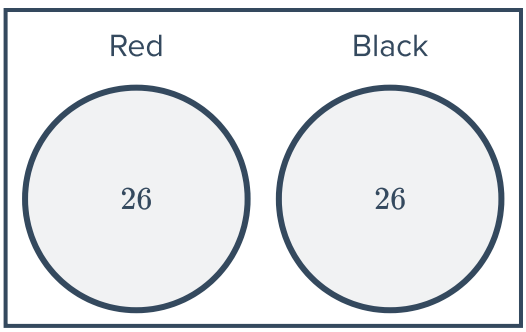
a Venn diagram 4

b Venn diagram 3

c Venn diagram 2

d Venn diagram 1

28



29

a $A \cap B = \{7, 8\}$

b $A \cup B = \{2, 4, 6, 7, 8, 10, 14\}$

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Venn diagrams (Extended)

30

a $A' = \{7, 14, 12, 10\}$

b $B' = \{4, 5, 1, 12, 10\}$

31

a $A = \{1, 3, 4, 5, 9, 10\}$

b $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

c $B' = \{1, 2, 9, 10, 11, 13\}$

32

a $A \cap B' = \{3, 4, 6\}$

b $(A \cup B)' = \{15, 60\}$

33

a Yes

b Yes

34

a $\{4, 5, 9\}$

b $\{1, 2, 3, 7, 9, 10, 11, 12, 13, 14\}$

c $\{4, 5\}$

d $\{3, 4, 5, 9\}$

e $\{1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

35

a Yes

b No

36

a 2 **b** 5 **c** 3 **d** 4

e 8 **f** 7 **g** 1 **h** 6

37

a

	Play Rugby League	Don't play Rugby League
Play Rugby Union	43	185
Don't play Rugby Union	179	146

b

i 179 **ii** 222 **iii** 228 **iv** 185

v 331 **vi** 325

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